						2/2023
Monday	–	6	13	20	27	
Tuesday	–	7	14	21	28	
Wednesday	1	8	15	22		
Thursday	2	9	16	23		
Friday	3	10	17	24		
Saturday	4	11	18	25		
Sunday	5	12	19	26		

06.week

Q 1:

$P(L_A) = 0.1$

$P(L_B) = 0.2$

(i) $P(L_A|L_B) = \dfrac{P(L_A \cap L_B)}{P(L_B)} = 0.35 \Rightarrow P(L_A \cap L_B) = 0.07$

(ii) $P(L_A \cup L_B) = P(L_A) + P(L_B) - P(L_A \cap L_B) = 0.23$

$P(L_A \cap L_B) = 0.07 \neq P(L_A)P(L_B)$. Hence not independent

c

$N = I_A + I_B$

(i) $E(N) = E(I_A) + E(I_B) = P(L_A) + P(L_B) = 0.3$

(ii) $Var(N) = Var(I_A) + Var(I_B) + 2.Cov(I_A, I_B)$

$\rightarrow Var(I_A) = P(L_A)(1 - P(L_A)) = 0.1 \times 0.9 = 0.09$

$\rightarrow Var(I_B) = P(L_B)(1 - P(L_B)) = 0.2 \times 0.8 = 0.16$

$\rightarrow Cov(I_A, I_B) = E(I_A I_B) - E(I_A)E(I_B)$

$= P(L_A \cap L_B) - P(L_A)P(L_B)$

$= 0.07 - 0.02 = 0.05$

d) If we assume naively independent,

$P_n(L_A \cap L_B) = P(L_A) \times P(L_B) = 0.02$

$P_{true}(L_A \cap L_B) = 0.07$

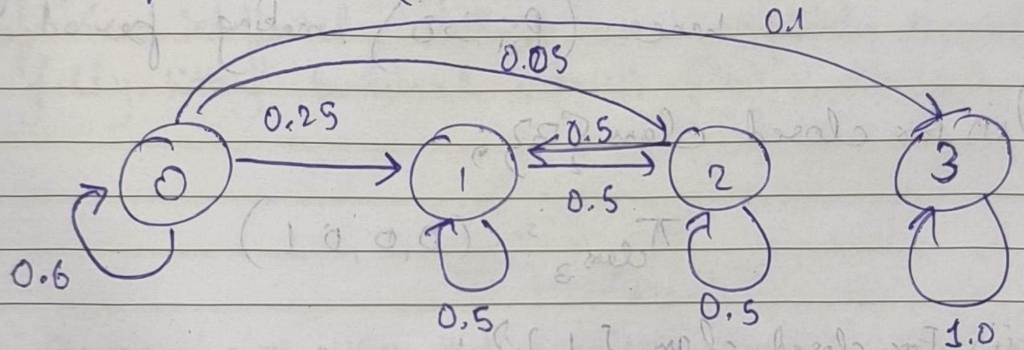
This underestimates the true joint prob. by a factor of

$\Rightarrow \dfrac{P_n(L_A \cap L_B)}{P_{true}(L_A \cap L_B)} = \dfrac{0.02}{0.07} = 2/7 \approx 0.2857$

(e) If systemic or asset-specific shocks cause asset losses to spike simultaneously, a naive model will severly underestimate the likelihood of extreme joint tail-risk events, leading to poor capital allocation, insufficient cash reserves and portfolio vulnerability during market crashes.

Q2.

(a) State space is $S = \{0, 1, 2, 3\}$



(b) A communicating class consists of ~~space~~ states that mutually communicate ($i \leftrightarrow j$)

(i) Class-1 : $\{0\}$, it can reach all states, but no state can return to 0, hence not closed.

(ii) Class-2 : $\{1, 2\}$, they transition back & forth mutually closed

(iii) Class-3 : $\{3\}$ Absorbing state, closed

(c) States 1, 2 & 3 are recurrent {belong to finite closed communicating class}

State 0 is transient {once left, system can never return}

(d) The chain is not reducible due to multiple ~~distinct communi~~ distinct communicating classes ($\{0\}, \{1,2\}, \{3\}$)

Aperiodicity – ~~State~~ Yes, every class is aperiodic because states 0, 1, 2, 3 all have self loops hence ($P_{ii} > 0$), making period 1.

(e) (i) For closed class $\{3\}$:

$$\pi_{class_3} = (0, 0, 0, 1)$$

(ii) For closed class $\{1,2\}$:

→ using sub matrix $P_{\{1,2\}} = \begin{pmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{pmatrix}$

→ solving $(\pi_1, \pi_2) P_{\{1,2\}} = (\pi_1, \pi_2)$ and $\pi_1 + \pi_2 = 1$

$= \pi_1/2 + \pi_2/2 = \pi_1 = 1$ $\quad \pi_1 = \pi_2 = 0.5$

→ if expanded to full state space

$$\pi_{class\{1,2\}} = (0 \;\; 0.5 \;\; 0.5 \;\; 0)$$

(f) Since there are 2 closed communicating classes, any convex combination of their stationary distributions is also a stationary distribution for the entire chain.

$\pi_{full} = \alpha(0 \;\; 0.5 \;\; 0.5 \;\; 0) + (1-\alpha)(0 \;\; 0 \;\; 0 \;\; 1)$ $\quad \alpha \in [0,1]$

$\pi_{full} = (0 \;\; 0.5\alpha \;\; 0.5\alpha \;\; 1-\alpha)$

~~(ii) Even when a theoretical stationary distributn is available, simulation is essential for analyzing path-dependent metrics. It allows risk managers to evaluate maximu~~

(i) Since class {1,2} is closed in corrected matrix, it represents a structural "distress trap". If a company's credit rating falls from investment grade {0} into speculative grade {1} or distressed state {2}, The model implies they will cycle back & forth b/w high-risk operations indefinitely without ever completely recovering or fully hitting terminal default {3}.

Q3.

(a) (i) Recurrence - implies that a market-making desk will safely return to a perfectly risk-neutral, zero inventory position ∞ many times over an ∞ horizon with absolute certainty.

(ii) Transience - implies that the market-making desk's inventory position will eventually diverge to ∞ and never return to 0 again, leading to unbounded inventory accumulation and uncontrollable capital risk exposure.

(b) A state i is recurrent if and only if the expected total no. of visits to State i starting from state i is ∞;

$$\sum_{n=0}^{\infty} p_{ii}^{(n)} = \infty$$ (conversely, if $\sum_{n=0}^{\infty} p_{ii}^{(n)} < \infty$, state is transient)

(c) For a simple symmetric random walk on $\mathbb{Z}^d$, the prob. of being back at the origin at step $2n$ asymptotically scales as:

$$p_{00}^{(2n)} \sim \frac{C_d}{n^{d/2}}$$

→ for $d = 1$: $p_{00}^{(2n)} \sim \frac{C_1}{n^{1/2}}$. Since $\sum_{n=1}^{\infty} \frac{1}{n^{1/2}}$ diverges, the walk is recurrent

→ for $d = 2$: $p_{00}^{(2n)} \sim \frac{C_2}{n}$. Since $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges, the walk is recurrent

→ for $d = 3$: $p_{00}^{(2n)} \sim \frac{C_3}{n^{3/2}}$. Since $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$ converges, walk is transient

(g) Simulation is fundamentally restricted to finite time horizon (n) and a finite no. of paths. It cannot prove analytical convergence properties because a path that hasn't returned within 10,000 steps might return at step 10^{10}, which cant be emperically validated within computational limits.

Q4.

(a) Irreducibility: The chain is irreducible because every state can transition to every other state in a finite no. of steps

Aperiodicity – Every diagonal element $(P_{ii}) > 0$
The presence of self loop ensures period of each state is 1. Thus, the chain is aperiod

(b) To solve $\pi P = \pi$ alongside $\sum \pi_i = 1$;

$$(\pi_0 \ \pi_1 \ \pi_2 \ \pi_3) \begin{pmatrix} 0.6 & 0.3 & 0.1 & 0 \\ 0.2 & 0.5 & 0.2 & 0.1 \\ 0.1 & 0.3 & 0.4 & 0.2 \\ 0.05 & 0.15 & 0.3 & 0.5 \end{pmatrix} = (\pi_0 \ \pi_1 \ \pi_2 \ \pi_3)$$

we'll evaluate this using python.

(c) Market Interpretation of π_i – The component π_i represents the unconditional, long-run invariant probability that the market will occupy regime i in any randomly selected week, independent of the initial starting state of the macroeconomy.

(j) Even when a theoretical stationary distribution is available, simulation remains essential for analyzing path-dependent metrics. It allows risk managers to evaluate maximum drawdown profiles, the distribution of consecutive down-weeks, tracking error distributions, and option pricing parameters that a static stationary distribution vector cannot capture.